

Abstract

Personnel matching is a pretty relevant topic in modern football strategy. While coaches have traditionally matched personnel based on player type (with the typical strategy being to add an extra defensive back for every receiver added to the field), offensive play callers have begun to exploit this strategy by utilizing heavy packages (12, 13, etc.) and passing at a high rate. This paper attempts to use a Bayesian Hierarchical Model to optimize defensive packages based on team, tendency, game context and offensive package. While the results are promising with serious improvements over an initial XGboost model, a package confounding issue still leaves some room to be desired for this model's deployment for pro football teams to use. With that being said, the promise this model's concept brings can help permanently influence how defense is played by implementing a prescriptive modeling approach to in game decision making traditionally dominated by coaching intuition.

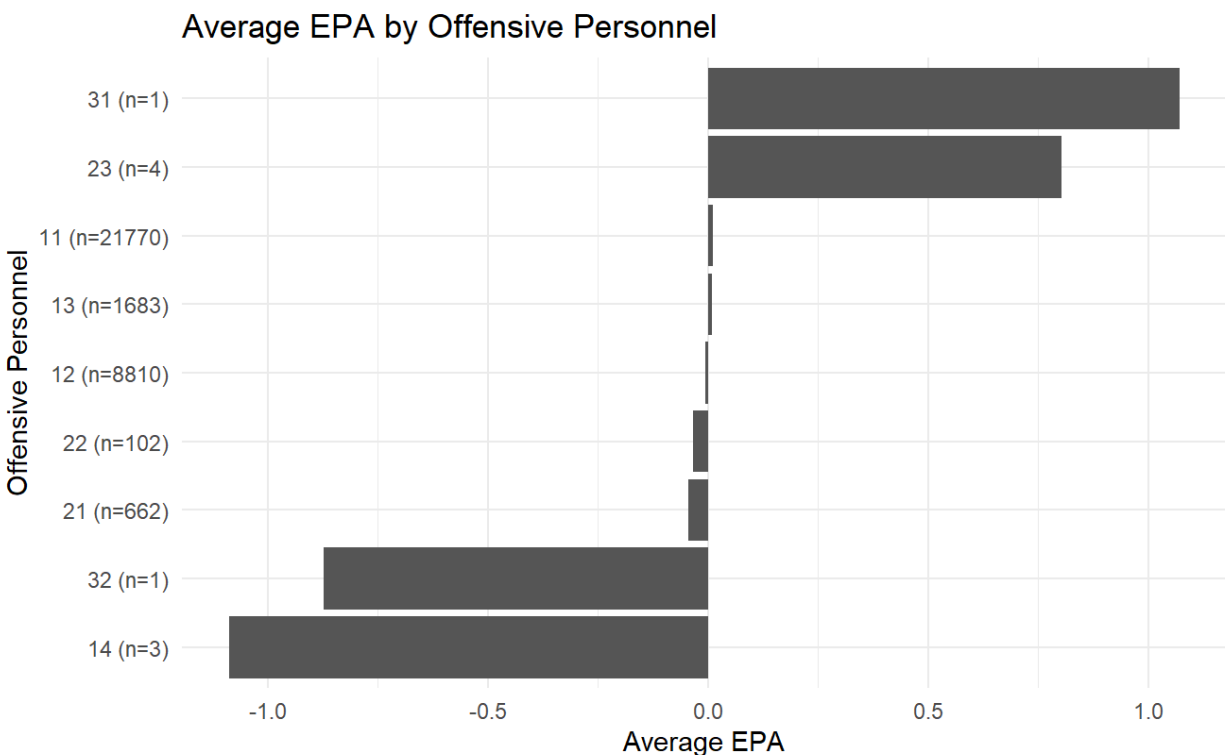
Related Work

There is plenty of literature that analyzes various aspects of coach decision making. Perhaps the most notable of these is [Ben Baldwin's 4th down model](#). This model analyzes decisions a coach can make on 4th down and based on its impact to win probability, recommends a specific decision. Most often those decisions consist of "kick a field goal, punt, or go for it". The reason this research is being addressed is because of the formatting. The simple "choose option A, B, or C," structure is extremely interpretable for coaches and fans alike which is optimal in the football research space. While more details will be addressed in its respective section, an ideal version of the model we seek to produce would have a similar structure with the options being, "base, nickel, or dime". The model will recommend one of these options based on what is expected to allow the lowest EPA on defense. [Norfolk, 2011](#) explores a similar avenue of research but keeps the main focus on optimal run pass splits. The model explores the interactions between choosing to "defend the run" versus "defending the pass" and explores the game theory that can arise from this sort of coach interaction. The paper concludes with the paradoxical view that teams that are most effective in passing should pass less and teams that are most effective in running should run less. While this is not the exact topic being explored in this report, it is tangential to the research we will conduct for one specific reason. Coaches are indirectly making this decision with their personnel selection (as explained in greater detail in the Background section). Coaches generally choose a heavier package on defense as a way of defending a more "run friendly offensive package" (Hence why coaches generally match 13 personnel with base). Finally, [Pitts, 2020](#) explores the impact of defensive decision makers and various factors that influence defensive effectiveness. The paper reaches the following conclusions. "We find that elite and very poor defensive coordinators and head coaches can have meaningful impacts on their teams' defensive performances. A team's defensive performance is shown to improve with increased defensive coordinator tenure. Similarly, defensive coordinators and head coaches who were NFL players contribute to improved defensive performance compared to coaches without experience as NFL players. In addition, a 3-4 defensive alignment is found to be associated with improved defensive performance compared to a 4-3 defensive alignment." ([Pitts, 2020](#)) The existing research demonstrates some interesting trends to look out for in this research project. First, usability is an important aspect of model effectiveness. There are many 4th down models that exist and it's entirely possible these models are more accurate than Baldwin's. I would make the case the reason Baldwin's model is the most popular is its usability and intuition. Because of this, arguably the most important part of an effective model is its usability. Second, there is such a thing as "optimal decision making" in terms of defending the run and pass. While it is built on assumptions, this can exist as a strategic framing device for defensive personnel choices. Finally, defensive coordinators matter. This is important because an important decision defensive coaches make every play call is the package to utilize. While I likely wouldn't have the capacity to contribute much in the schematic department, due to the hundreds of millions of defensive coverage combinations

coaches can utilize. I can contribute to a team's decision making by finding the mathematically optimal personnel decision based on historical data. It is significantly easier to recommend one of three choices as opposed to one in millions.

Data Overview

The data for this project is in the popular R package [nflreadR](#). This package contains EPA, personnel, yardline, team, and football statistic data. The personnel data is in a different format than the previously established xx number format but that format can be derived by counting the number of running backs and tight ends on offense and the number of total defensive backs on defense. While there are many variations of defensive schemes and packages, for simplicity sake we will group each into one of the three following based purely on the number of defensive backs on the field. Base will be when four defense backs are on the field, Nickel will be when five defensive backs are on the field, and Dime will be when six defensive backs are on the field. The model analyzed play by play data for the 2023 - 2025 NFL seasons. The training data will be the 2023 through the first half of the 2025 NFL season with “recency weights” set on each season. 2025 data is weighted at 1.0 given that it's the most recent and relevant season, 2024 data is weighted at 0.8, and 2023 is weighted at 0.6. These weights can likely be improved with an exponential decay function or something more objective but for now the 0.2 decrease each season is solid enough. The model was then trained and tested on the second half of the 2025 season (weeks 10-18). This split was done because of the temporal importance of the data. Testing on past seasons would likely lead to data leakage and its important to note that teams are continuously evolving. Strategies from one season may be different from previous seasons and as a result, choosing to test on only the most recent season we have data on was a choice made to preserve the integrity of the model. Initial analysis of the data results in the following chart. This chart was created by grouping each play by its personnel grouping and taking the mean EPA for that grouping. These plays were from the 2025 season.



- Figure 1

The personnel groupings with the widest swings were unsurprisingly the groupings with the lowest sample size. The personnel with the highest sample sizes by far were 11, 12 and 13 personnel which is the largest reason they are the study of this research paper. One of the more interesting trends to notice is that there is not a serious EPA advantage from any specific personnel grouping. While 11 personnel is the highest in terms of average EPA, the range between 11 and 12 personnel is less than 0.1 EPA meaning team strategy game context and matchup are key to both being successful at both utilizing and stopping these personnel groupings.

Methods

XGBoost

The first model used in this research was a baseline XGboost. The model is designed to predict EPA and to do that, the model was broken into three parts to more effectively estimate the causal impact of a defensive personnel. The complete formula for the three models is as follows.

$$E[EPA | package] = P(pass | package) \times E[EPA | pass, package] + P(run | package) \times E[EPA | run, package]$$

The first model used the following features to predict the probability of a pass. , "ydstogo", "yardline_100", "score_differential", "quarter_seconds_remaining", "half_seconds_remaining", "game_seconds_remaining", "posteam_enc", "defteam_enc", "off_pers_enc", "def_pers_enc"

Down - 1st, 2nd, 3rd, or 4th down? Teams only get four chances to get a first down making this a key feature in predicting passes. Generally, the higher the down the higher the pass rate.

Yards to go - How many yards are necessary for getting a first dow?. This is critical due to this example. 3rd and 10 the offense must get 10 yards in that single play, making a pass highly likely. 3rd and 1 and the offense only has to get 1 yard in that play, making a run more likely.

yardline_100 - How many yards are necessary to score? 99 would mean the team is on the verge of scoring, 1 would mean the team has 99 yards to go to score.

Score Differential - How much is the offense leading by? Pass rates change a lot depending on the situation especially because a team who possesses the ball with a lead now has two primary goals, maintain an efficient offense and eat up as much time on the clock as possible. Often these goals are somewhat contradictory considering running plays are more effective at eating up clock while also being lower in EPA per play.

Quarter seconds, half seconds, game time remaining - All of these will be lumped together. Teams operate differently depending on time remaining, if a team is trailing while there is little time remaining, expect their pass rate to be extremely high.

Team - Who is on offense and possesses the ball? This will come later when analyzing EPA outcomes but it is important to get historic rates, some teams prefer to pass at higher rates while others pass at lower rates, this feature allows the model to learn team tendency.

Defense team - Who is on defense? This is to determine who is on defense and will later become more important when analyzing EPA. This is also the team we are making prescriptions for with our model.

Offensive Personnel - What personnel is the offense in? This is used to train the model on what team * model combination is most efficient and what the expected pass rate is given the offensive personnel * defensive personnel interaction.

Defensive Personnel - What personnel is the defense in? This is used to train the model on what team * personnel combination is most efficient and what the expected pass rate is given the offensive personnel * defensive personnel interaction.

The other two models share many of the same features so those will not be repeated. The second and third models are the expected EPA given the play is a pass and the expected EPA given that the play is a run. This model uses all of the previously named features with the addition of two features.

Pass EPA - Expected points added (or yards with context) for the passing play. It is worth noting passes as inherently more varied in their distribution thus leading to a wider range of outcomes, making it more difficult to predict.

Run EPA - Expected points added for the running play.

The model performance is below in figure 2.

Model Performance — 2025 Weeks 10–18 Test Set	
Metric	Value
Run EPA RMSE	0.9934
Pass EPA RMSE	1.6152
Play type accuracy	69.49%
Play Type ROC-AUC	0.7687

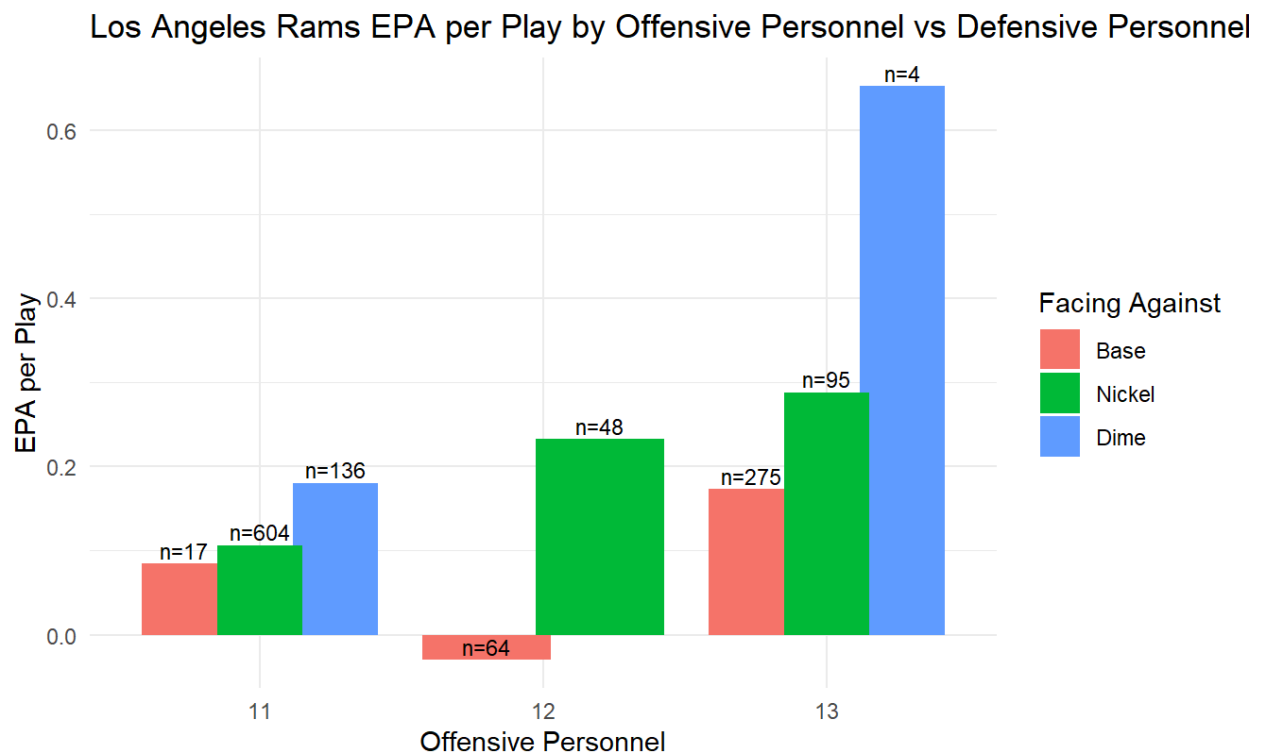
- Figure 2

It is unsurprising to see a higher RMSE for pass EPA given the natural variance of passing plays. Turnovers are more likely, sacks are possible and explosive plays are significantly higher. The important thing to note is the play

type accuracy. 70% is a pretty solid improvement over the 57% baseline (roughly 57% of plays are pass plays) but there is definitely room to grow. Play type accuracy is very important in the context of this project because of the potential for compounding errors. A low play type accuracy and a high EPA RMSE compounds the inaccuracy of the model, making it causally sound but leading to a drop in accuracy. Given the modest but promising results, let's discuss the decision to transition to a Bayes Hierarchical Model.

Bayes Model

The NFL has no shortage of sample size for many testable questions. Unfortunately for this project, there are many personnel matchups that are simply rare. Let's take Dime against 13 personnel for example. The Los Angeles Rams faced Dime personnel while in 13 personnel a grand total of 4 times in the 2025 season. Figure 3 shows the personnel matchups. Due to that rarity, it is difficult to gain much insight from using traditional frequentist modeling to tackle this problem.



- Figure 3

Bayesian models use a concept known as partial pooling to allow samples to influence other samples. In a case like this, the model can analyze a defense's nickel and base performance and use it to influence the expected EPA when in dime. This concept can be expanded to other aspects of the data including team sampling (i.e. if there is limited data on the Vikings playing against 13 personnel while in Dime, the model will pull information from the rest of the league to influence expected EPA returns for runs and passes)

Results

The model saw a significant improvement in accuracy from the XGBoost baseline. Figure 4 shows the comparison between the two models.

Model Performance Comparison — 2025 Weeks 10–18 Test Set		
Metric	XGBoost	Bayesian
Play type accuracy	69.48%	76.39%
AUC-ROC	0.7683	0.8252
Brier score	N/A	0.1678
Brier skill score	N/A	31.8% better than naive
Pass EPA RMSE	1.6152	1.5550
Pass EPA CRPS	N/A	0.8163
Run EPA RMSE	0.9930	0.7664
Run EPA CRPS	N/A	0.3689

- Figure 4

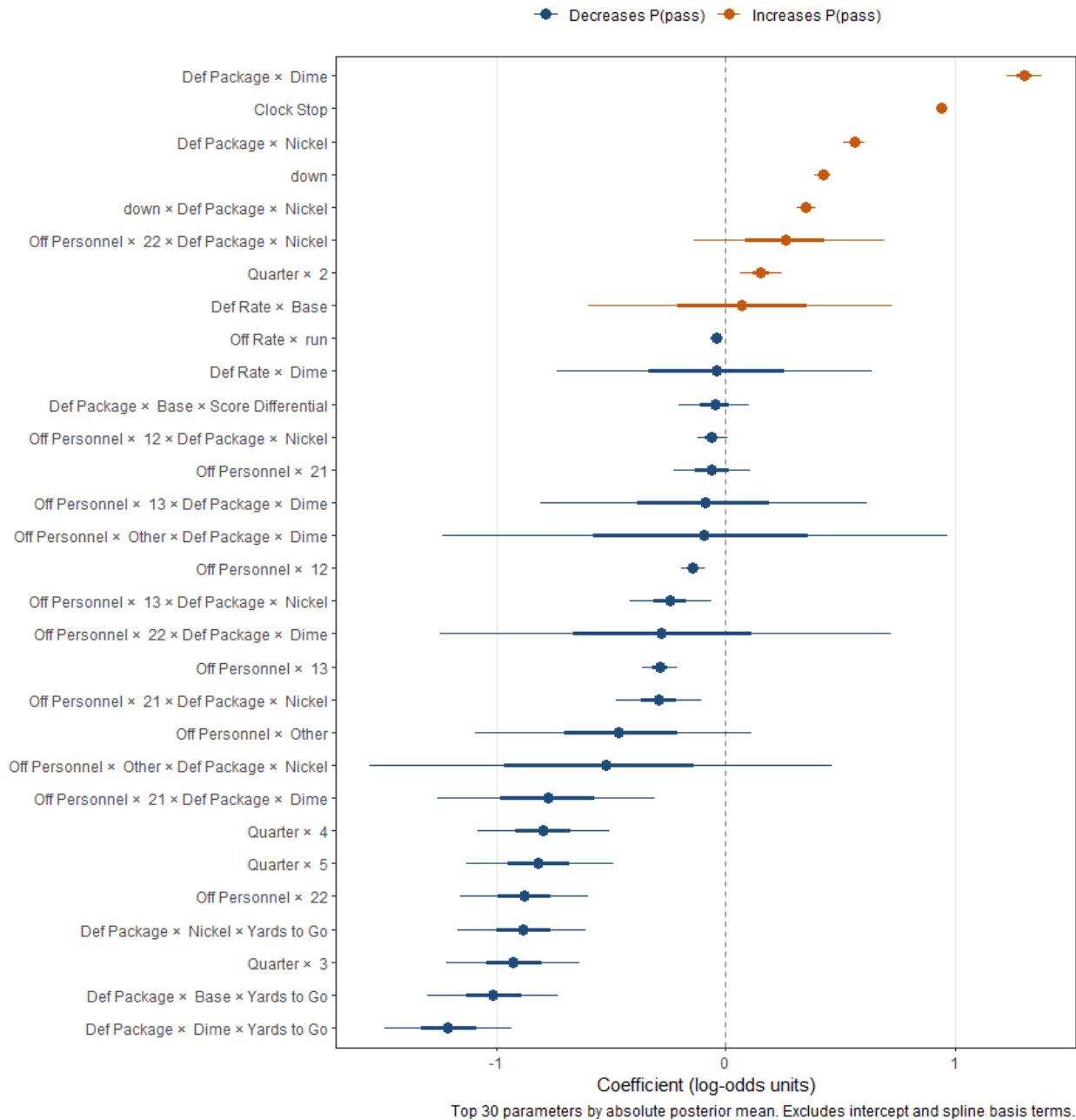
A 7% improvement in play type accuracy is extremely promising especially given the importance of having a high accuracy as explained in the methods section. The Bayes model showcases an improvement in every shared metric. The table also shows some metrics unique to the Bayes model. Briar score formula is as follows

Brier = average of (predicted probability - actual outcome)²

This is especially important for a Bayes model which is heavily reliant on grading the model's confidence. Figures 5, 6 and 7 show the features that most heavily impact the model using fixed effect coefficients (think of them as the Bayes version of SHAP graphs)

Stage 1: Fixed Effect Coefficients — $P(\text{pass} \mid \text{situation}, \text{def_package})$

Log-odds scale. Points = posterior mean. Thick bar = 50% CI. Thin bar = 90% CI.



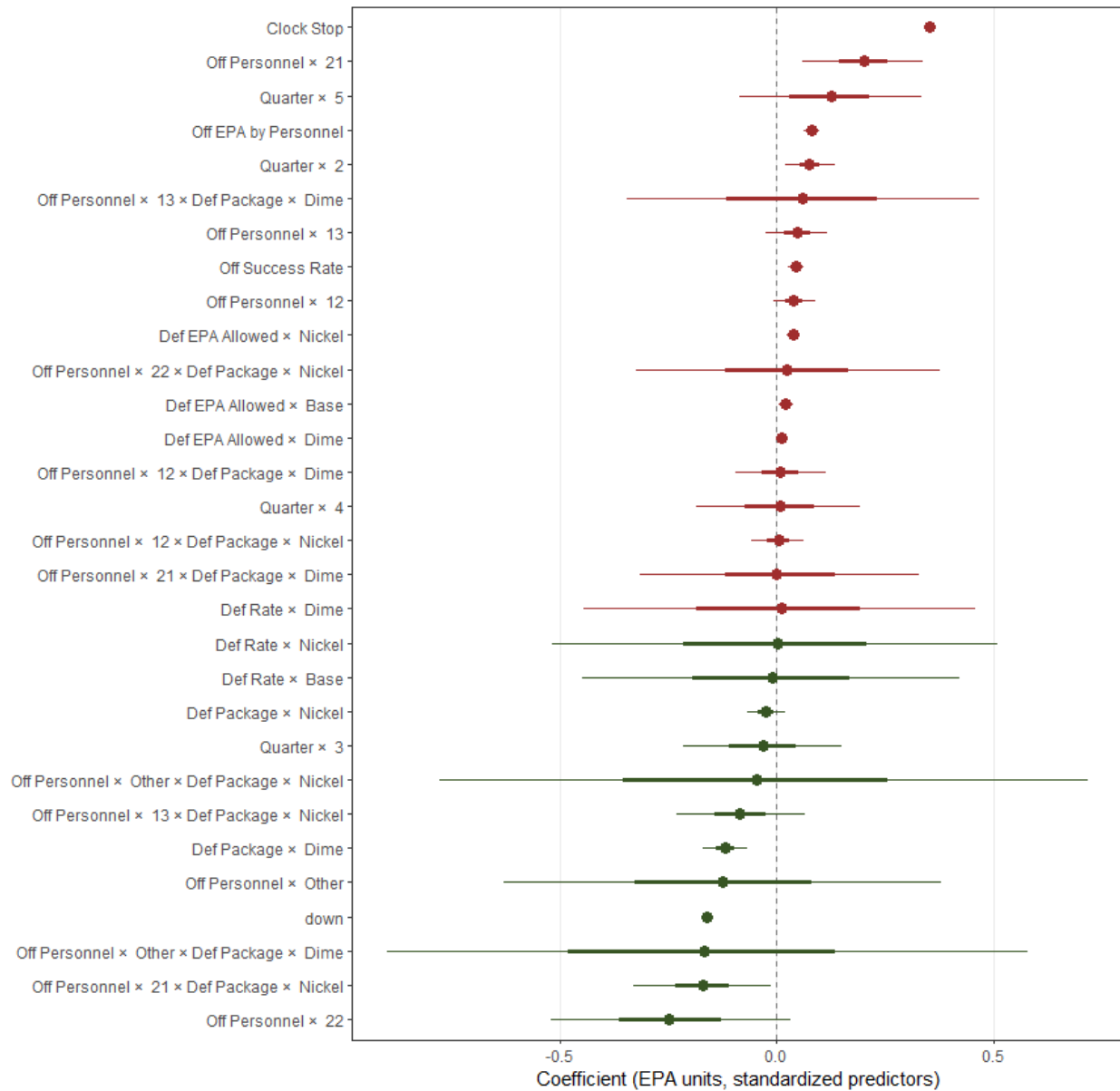
- Figure 5

Dime is by far the greatest predictor of the probability of a pass. This makes sense given the fact that defenses typically only use dime when in obvious passing situations. This feature, while boosting the accuracy, does contribute to a confounding issue that affects the prescriptive part of the model that will be discussed later on. Other features such as down, quarter and clock stop are features that influence the likelihood of a pass.

Stage 2a: Fixed Effect Coefficients — $E[EPA | \text{pass}]$

EPA units. Points = posterior mean. Thick bar = 50% CI. Thin bar = 90% CI.

—●— Decreases Pass EPA —●— Increases Pass EPA



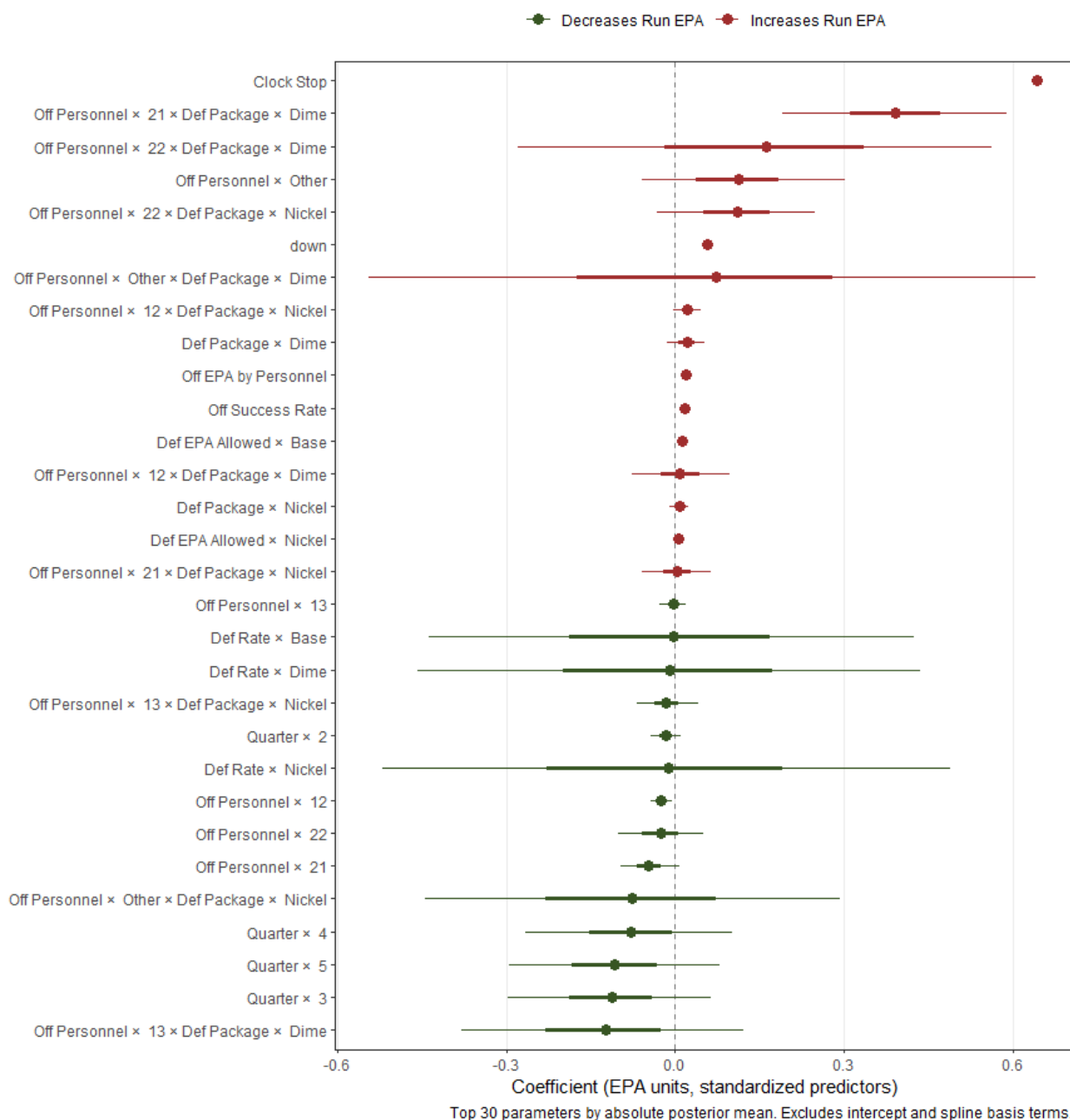
Top 30 parameters by absolute posterior mean. Excludes intercept and spline basis terms.

- Figure 6

Clock stop, 21 personnel, and offensive success rate are all features that heavily influence expected EPA on passing plays.

Stage 2b: Fixed Effect Coefficients — E[EPA | run]

EPA units. Points = posterior mean. Thick bar = 50% CI. Thin bar = 90% CI.



- Figure 7

Interaction terms are particularly interesting here. The model understands that dime performs poorly against heavier packages in terms of its ability to defend the run. Down is another important influence on run EPA.

Actions

The main purpose of this project is to optimize defensive matchups. In order to achieve that the model simulates the expected EPA allowed for each defensive personnel grouping. By receiving the following inputs, down, distance, time left in half, time left in game, offensive team, defensive team and offensive personnel, the model will return the following outputs. Expected pass probability, posterior EPA distribution and a personnel recommendation that allows the lowest expected EPA. Figure 8 shows what the model's app and recommendations look like.



NFL Defensive Personnel Optimizer

Bayesian Hierarchical Model — 2025 Season — Built by Asa Arnold

MATCHUP

Offense



Los Angeles Rams

Defense



Minnesota Vikings

GAME SITUATION

Down

1st

Distance (yds)

10

Yards to End Zone

30

Quarter

Q4

Score Differential (offense POV)

7

Positive = offense leading. E.g. +7 = offense up by 7.

Game Secs Remaining

395

Half Secs Remaining

395

OFFENSIVE PERSONNEL

Personnel Group

13 — 1 RB 3 TE 1 WR

► Get Recommendation



Los Angeles Rams

vs



Minnesota Vikings

1st & 10 • 30 yds to end zone • Q4 • +7 score • 13 personnel

6:35 remaining in game

Dime

★★ Recommend

52.7% probability this is the optimal package

KEY NUMBERS

EXPECTED EPA

-0.327

lower = better for D

80% INTERVAL

[-0.519, -0.135]

posterior uncertainty

P(PASS) IF SHOWN

39.5%

offense reaction estimate

EDGE VS #2

+0.028

EPA over second-best

ALL PACKAGE COMPARISON

Dime

RECOMMENDED

-0.327 EPA

[-0.519, -0.135]

39.5% pass

P(best): 52.7%

Nickel

-0.298 EPA

[-0.365, -0.231]

17.7% pass

P(best): 27.3%

Base

-0.290 EPA

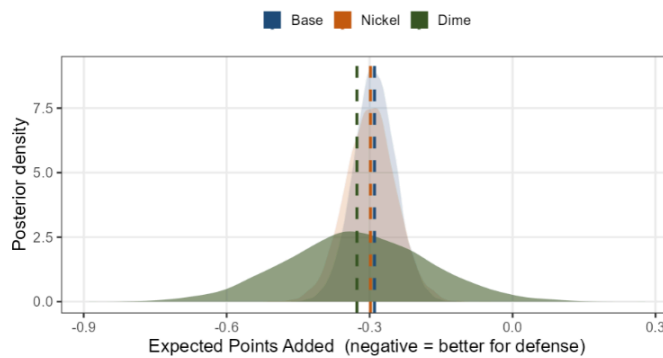
[-0.346, -0.234]

16.7% pass

P(best): 20.0%

EPA POSTERIOR DISTRIBUTIONS

Highlighted = recommended package. Dashed lines = posterior means.



- Figure 8

While the model shows immense promise, there are a few things worth noting that prevent the model from being “deployment ready”. The first of which is the drastic underestimation of EPA. The Los Angeles Rams are a top 5 offense in football and average more than 0.2 EPA per play from 13 personnel, because of that, an expected EPA of -0.327 is quite low. Second, the expected pass probability is not logically coherent. Dime in neutral situations (like a 1st and 10) should increase the offense’s likelihood of running, not decrease it. The reasoning for this is simple. Dime is objectively worse against the run by both the model’s standards and traditional coaching intuition. A rational coach would increase their likelihood of running and yet the model projects a higher chance the offense will pass when the defense is in dime. This is the result of training the $P(\text{pass} \mid \text{package})$ aspect of the model on observational data. The reality is, teams very rarely implement dime defense in neutral situations. The only time dime is implemented is typically in obvious passing situations. The model sees this relationship and assumes game situation and the dime package itself are causing a high likelihood of passing when in reality it's the game situation and the high likelihood of passing that causes the dime coverage. Ways of fixing this component of the overall model will be discussed in conclusion and future work. For now, the model is highly effective at a few things. The first of which is understanding the importance of stopping the pass. In several instances, the model prefers dime or nickel (two packages with a heavier emphasis on stopping the pass) over base which I think shows some promise for the analytics crowd. Second, it has an understanding of a team’s effectiveness from a certain personnel package. Let’s look at a model output with identical inputs to Figure 8 with one key change. Instead of the Rams being the team on offense, the Raiders will be the team on offense. This team was chosen because they ranked 32nd in total EPA from 13 personnel ([courtesy of SumerSports](#)). Figure 9 shows the difference in the output.

NFL Defensive Personnel Optimizer

Bayesian Hierarchical Model — 2025 Season — Built by Asa Arnold

MATCHUP

Offense



Las Vegas Raiders

Defense



Minnesota Vikings

GAME SITUATION

Down

1st

Distance (yds)

10

Yards to End Zone

30

Quarter

Q4

Score Differential (offense POV)

7

Positive = offense leading. E.g. +7 = offense up by 7.

Game Secs Remaining

395

Half Secs Remaining

395

OFFENSIVE PERSONNEL

Personnel Group

13 — 1 RB 3 TE 1 WR

► Get Recommendation



Las Vegas Raiders

VS



Minnesota Vikings

1st & 10 • 30 yds to end zone • Q4 • +7 score • 13 personnel
6:35 remaining in game

Dime

★★ Recommend

78.4% probability this is the optimal package

KEY NUMBERS

EXPECTED EPA

-0.742

lower = better for D

80% INTERVAL

[-0.934, -0.553]

posterior uncertainty

P(PASS) IF SHOWN

35.7%

offense reaction estimate

EDGE VS #2

+0.135

EPA over second-best

ALL PACKAGE COMPARISON

Dime

RECOMMENDED

-0.742 EPA
[-0.934, -0.553]

35.7% pass
P(best): 78.4%

Nickel

-0.607 EPA
[-0.678, -0.536]

15.4% pass
P(best): 13.2%

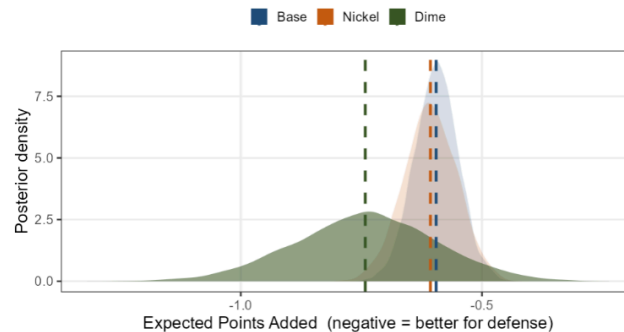
Base

-0.595 EPA
[-0.653, -0.537]

14.5% pass
P(best): 8.4%

EPA POSTERIOR DISTRIBUTIONS

Highlighted = recommended package. Dashed lines = posterior means.



- Figure 9

Notice the increased confidence in the model and this is likely for a few reasons. First, the Raiders were 28th in EPA per rush. The model could more confidently recommend dime given that running became less of a threat. Second, the expected EPA is significantly lower largely because the defense is facing a less effective offense. These outputs show immense potential for both this current iteration of the model and the general approach of using a Bayesian Hierarchical Model to model out this research question. In the model's ideal form, a coach's assistant will input the situation and the model will generate a recommendation. The most important part of this recommendation is the model's confidence, a 52% confidence recommendation would allow the coach to feel less pressure to conform to the model while a 70% confidence recommendation will give more of a justification in following the model. This could heavily influence the game especially in defending teams like the Rams who made it a point to exploit heavy personnel usage.

Conclusion and Future Work

The largest opportunity for improvement is addressing the dime confounding issue previously discussed. The model in its current state cannot effectively distinguish how personnel causes the likelihood of a run or pass. There are a few potential methods to improve this process. Probably the most rigorous but successful option would be to use game theory to determine the likelihood of a run or pass. Instead of using observational data, the model could take a "what should coaches do?" approach as opposed to the current iteration which is taking a "what would coaches do?" approach. The framework to do something like this would be to utilize a Nash equilibrium that uses the $E[\text{EPA} \mid \text{pass, package}]$ and $E[\text{EPA} \mid \text{run, package}]$ generated by the model to determine the offense's likelihood of running.

Second, the model using EPA as its output is ideal for most in-game situations but does not always capture what a coach is attempting to do. In cases where the team has the lead, coaches are often more focused on draining the clock than they are generating high EPA plays. Because of that, using win probability or success rate could be another option to improve the model.

Finally, the model views each play in complete isolation. This means it is not familiar with any of the in-game context. Perhaps the offensive play caller purposely ran the ball from 13 personnel in order to generate a play action pass. The model views each play in complete isolation which is somewhat of a strength (may not fall for play action setups) and a weakness (may not understand how to adjust to several unsuccessful defensive plays). A more effective version of this model could repeatedly train and adjust and place the heaviest weight on recent weeks or even recent plays that occurred in the game. The model could also have an "in game context" feature which views how many times a coach ran from a specific personnel already to learn tendencies.

While there are several faults in the current iteration of the model, the framework has proven to be promising. Research like this can have a significant impact on the way the game is played.

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